# CAPSTONE PROJECT REPORT

## *E-Commerce Sales & Customer Insights Dashboard*

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| **Batch/Program** | **2023-2027\|B.Tech-Information Technology** |
| **College** | **Kalinga Institute of Industrial Technology (KIIT)** |
| **Project Title** | **E-Commerce Sales & Customer Insights Dashboard** |
| **Tools Used** | **Microsoft Power-BI** |
| **Submission Date** | **April 19,2026** |

Kalinga Institute of Industrial Technology

Department of Information Technology | B.Tech 2023–2027

# Problem Statement

Modern e-commerce businesses generate large volumes of transactional data, including sales records, customer information, and product details. However, this data often remains underutilized due to the lack of effective analytical and visualization tools.

Without a structured approach, it becomes difficult to:

- Identify sales trends over time
- Understand customer purchasing behavior
- Evaluate product performance
- Compare regional contributions to revenue

As a result, decision-making becomes inefficient and reactive rather than data-driven.

This project aims to address these challenges by developing an interactive Power BI dashboard that transforms raw data into meaningful insights. The dashboard enables users to analyze sales performance, customer behavior, and regional trends in a clear and actionable manner.

# 2. Objectives of the Project

The main objectives of this project are:

- To design an interactive dashboard for analyzing sales data
- To track key business metrics such as revenue, orders, and customers
- To analyze customer behavior and purchasing patterns
- To identify top-performing products and regions
- To provide data-driven insights for better decision-making

# 3. Solution & Features

The project is implemented using Power BI and consists of three interactive dashboard pages:

## Page 1: Executive Overview

- Displays key KPIs:
    - Total Revenue
    - Average Order Value
    - Total Orders
    - Total Customers
- Monthly revenue trend analysis
- Revenue distribution by country
- Top products contributing to revenue
- Key business insights section

## Page 2: Customer Behavior & Insights

- Customer growth trend over time
- Customer distribution by country
- Identification of top customers
- Quantity distribution analysis (purchase behavior)
- Insights on customer purchasing patterns

## Page 3: Customer Loyalty & Value Insights

- Average orders per customer
- Revenue generated per customer
- Repeat customers and repeat rate
- Customer order distribution
- Revenue contribution by country
- Business insights and recommendations

## 4. Dashboard Screenshots

### Page 1 - Executive Overview

Page 2 - Customer Behavior & Insights

Page 3 - Customer Loyalty & Value Insights

# 5. Technology Stack

The following tools and technologies were used:

- **Power BI Desktop** – Dashboard creation and visualization
- **Power Query** – Data cleaning and transformation
- **DAX (Data Analysis Expressions)** – Calculations and measures
- **CSV Dataset (Online Retail Dataset)** – Data source

# 6. Data Preprocessing

Before building the dashboard, the dataset was cleaned and prepared using Power Query:

- Removed null values from important columns
- Filtered out negative and zero quantities
- Removed invalid transactions (e.g., canceled orders)
- Converted data types appropriately
- Created calculated columns for better analysis

This ensured data accuracy and improved dashboard performance

# 7. Key Measures (DAX)

The following measures were created:

- **Total Revenue** = Quantity × Unit Price
- **Total Orders** = DISTINCTCOUNT(InvoiceNo)
- **Total Customers** = DISTINCTCOUNT(CustomerID)
- **Average Order Value** = Total Revenue / Total Orders
- **Revenue per Customer** = Total Revenue / Total Customers
- **Repeat Customers** = Customers with more than one order
- **Repeat Rate** = Repeat Customers / Total Customers

# 8. Key Insights

- The United Kingdom contributes the majority of total revenue
- Sales show a consistent upward trend over time
- A small number of products dominate overall revenue
- Most transactions involve low quantities, indicating frequent small purchases
- A small group of customers generates a large portion of revenue
- Customer activity shows steady growth across months

# 9. Business Recommendations

- Focus marketing efforts on high-performing regions (e.g., UK)
- Introduce loyalty programs to retain high-value customers
- Target low-frequency customers with personalized campaigns
- Promote top-performing products to maximize revenue
- Improve engagement strategies to increase repeat purchases

# 10. Unique Features

- Interactive multi-page dashboard
- Use of slicers for dynamic filtering
- Clear and user-friendly design
- Business-focused insights and storytelling
- Combination of sales and customer analytics

# 11. Future Enhancements

- Integration with real-time data sources
- Use of machine learning for predictive analysis
- Advanced customer segmentation techniques
- Automated reporting and alerts
- Enhanced dashboard customization

# 12. Conclusion

This project demonstrates how Power BI can be used to transform raw data into meaningful business insights. The dashboard provides a comprehensive view of sales performance, customer behavior, and regional trends.

It enables businesses to make informed decisions, improve customer engagement, and optimize revenue strategies. Overall, the project highlights the importance of data visualization in modern data-driven environments.